

(A) Gate pipeline

Acoustic detections (OrcaHello)

L0 QC quarantine

PSTH kernel fit
 $\log \lambda = b_0 + \sum_k \beta_k \phi_k + \log E$ (E1)

Phase-shuffled null test (E2)

IC: no signal

Time-rescaling GoF (E3)

IC: GoF failed

Deviance skill S_D (E5)

IC: skill negative

PIT calibration (E6)

IC: miscalibrated

Effective confidence
 $c_{\text{eff}} = c_{\text{raw}} \cdot \prod_k g_k$ (E4)

Human promotion
 (DynamoDB)

Failing a gate surfaces an integrity condition alongside the metric; it does not suppress the forecast. Human promotion is the final gate.

metric
 $c_{\text{eff}}(\text{cell}) = 0\%$ (not yet promoted)

claim

C: Claim
 “lunar kernel beats null”
 gate_citation

validated_by

M: Method
 fetch_gates (T0 skill)
 skill_invocation step

evaluated_by

X: Experiment
 Level 1 PSTH (illustrative $p = 0.002$)
 repr_id, run_id

derived_from

D: Data
 acoustic candidates, QC mix
 provenance_citation

grounded_in

R: Research
 methodology.tex kernel spec
 artifact_citation

Left branch only. Each $C \rightarrow M \rightarrow X \rightarrow D \rightarrow R$ chain is rendered live from the interaction step log (interaction_steps JSONB). Scenario 4 injects this chain as grounding context, eliminating R_{uncited} .

provenance graph $G = (V, E)$

no signal (UI example)
 Claims without artifact
 render this badge.
 Absent from Scenario 4:
 step-log binds all claims.

supports